

# Adapting Small Vision-Language Models (<1B) for Document Understanding

---

## A systematic evaluation and improvement strategy

Author: Sangbum Choi · Date: 2026-06-17

---

## 0. Executive summary

---

Small VLMs (sub-1B parameters) are attractive for document understanding because documents are high-volume, privacy-sensitive, and increasingly processed on-device or at the edge — exactly where a 3B+ model is too expensive. But their document capabilities are under-characterised: most public leaderboards focus on 3B–70B models.

This report (1) selects eight sub-1B VLMs spanning generalists, edge models and OCR specialists; (2) defines a document-understanding benchmark suite anchored on **DocVQA**, **InfoVQA**, **ChartQA** and **OCRBench**, plus a **custom robustness probe**; (3) evaluates them with metrics that go beyond accuracy — **ANLS / relaxed accuracy / OCRBench score**, plus **calibration (ECE)** and **robustness retention**; and (4) turns the resulting gap analysis into a concrete, literature-grounded improvement plan.

**Headline finding (from published reference figures, to be reproduced with the included pipeline):** the **InternVL family at ~0.9B (InternVL2.5-1B / InternVL3-1B)** is the strongest document generalist in this size class — DocVQA  $\approx$  82–85 ANLS, OCRBench  $\approx$  785–790, ChartQA  $\approx$  75 — yet it has a **clear, consistent weak spot on InfoVQA ( $\approx$  54–56)**: multi-element, layout-and-numeric reasoning over infographics. This — not raw text recognition — is where the best small model falls short, and it is the target of the improvement strategy in Part 2.

**Reproducibility note.** This environment has **no GPU**, so the report does not present numbers I ran myself as if they were measured here. The **evaluation pipeline is complete, tested, and runnable** (CPU smoke-tested end-to-end; metric code unit-tested); the GPU runs are intended for a free Colab/Kaggle T4, for which exact commands are provided. Where I cite scores, they are **published reference figures** (model papers / cards) with explicit source and caveat, and the comparison table clearly separates "published reference" from "reproduced".

---

## Part 1 — Evaluation

### 1. Model selection criteria

I select for **architectural diversity within the <1B budget**, so the comparison explains why one design wins, not just which wins. Three archetypes are represented:

| # | Model                       | Params | Vision encoder + LM                                       | Archetype                | Why included                                               |
|---|-----------------------------|--------|-----------------------------------------------------------|--------------------------|------------------------------------------------------------|
| 1 | <b>InternVL2.5-1B</b>       | ~0.94B | InternViT-300M + Qwen2.5-0.5B, dynamic tiling             | Document generalist      | SoTA-for-size doc/OCR; first-class in VLMEvalKit           |
| 2 | <b>InternVL3-1B</b>         | ~0.94B | InternViT-300M + Qwen2.5-0.5B, native multimodal pretrain | Document generalist      | Newer pretraining recipe; head-to-head with 2.5            |
| 3 | <b>SmolVLM-500M</b>         | 0.5B   | SigLIP + SmoLLM2, Idefics3, heavy token compression       | Edge                     | Edge-deployment anchor; trained incl. Docmatix             |
| 4 | <b>SmolVLM-256M</b>         | 0.26B  | SigLIP + SmoLLM2                                          | Extreme edge             | Smallest serious VLM; floor of the size/quality curve      |
| 5 | <b>LLaVA-OneVision-0.5B</b> | ~0.9B  | SigLIP-SO400M + Qwen2-0.5B, AnyRes                        | General VLM              | "General-purpose small VLM" baseline (not doc-specialised) |
| 6 | <b>GOT-OCR2.0</b>           | ~0.58B | ~80M ViT + Qwen-0.5B decoder                              | OCR specialist           | Pure transcription; tests "OCR ≠ document QA"              |
| 7 | <b>Florence-2-large</b>     | ~0.77B | DaViT + BART enc-dec, task tokens                         | OCR/detection specialist | Unified task-token model; no free-form VQA                 |
| 8 | <b>PaddleOCR-VL-0.9B</b>    | ~0.9B  | NaViT-style encoder + ERNIE-4.5-0.3B                      | Doc-parsing specialist   | Newest, most document-specialised release                  |

## Rationale highlights

- **Why InternVL ×2.** It is the consensus strongest small document model, and including both 2.5 and 3 isolates the effect of the pretraining recipe at fixed architecture/size — directly relevant to "pretraining data" in the selection criteria.
- **Why two SmolVLM sizes.** They trace the size/quality frontier (256M→500M) and represent the edge-deployment use case the task emphasises; SmolVLM's training mix includes **Docmatix**, a large document instruction set, making it the strongest small-and-edge document model and a fair foil to InternVL.
- **Why a general baseline (LLaVA-OV-0.5B).** To quantify the premium document-specialised pretraining buys over a strong but general small VLM at the same ~0.9B size.
- **Why two OCR specialists (GOT, Florence-2) and a parser (PaddleOCR-VL).** Document understanding is not OCR. GOT/Florence-2 transcribe but cannot answer questions; including them makes the **"recognition vs. reasoning" gap measurable** rather than asserted, and PaddleOCR-VL tests whether a parsing-first design transfers to QA.

**Extended candidate pool (added after review).** The harness also registers six further sub-1B open-weight models so the comparison spans the current field, including a version ablation of the leading document specialist:

| Model                   | Params | Components                  | Note                                                                                            |
|-------------------------|--------|-----------------------------|-------------------------------------------------------------------------------------------------|
| <b>PaddleOCR-VL-1.5</b> | ~0.9B  | NaViT + ERNIE-4.5-0.3B      | v1.5: +polygon localization / text-spotting / seal; OmniDocBench v1.5 ≈ <b>94.5</b> (SOTA-tiny) |
| <b>InternVL2-1B</b>     | ~0.94B | InternViT-300M + Qwen2-0.5B | older 1B; DocVQA 81.7 / OCRBench 754 — recipe baseline vs 2.5/3                                 |
| <b>Ovis2-1B</b>         | ~1.0B  | AIMv2-large + Qwen2.5-0.5B  | structural visual-text embedding; strong OCR (OCRBench ≈ 89/100)                                |

| Model                         | Params | Components                  | Note                                                        |
|-------------------------------|--------|-----------------------------|-------------------------------------------------------------|
| <b>H2OVL-Mississippi-0.8B</b> | 0.8B   | InternVL-style + H2O-Danube | OCR/doc specialist (19M pairs); OCRBench 751                |
| <b>SmolDocling-256M</b>       | 0.26B  | SmolVLM-256M base           | smallest true document specialist; emits structured DocTags |
| <b>Florence-2-base</b>        | 0.23B  | DaViT + BART enc-dec        | smaller task-token sibling of Florence-2-large              |

Models documented but **not registered**: InternVL3.5-1B (~1.1B, borderline over budget), Moondream-0.5B (license unconfirmed via gated card), and the seq2seq specialists Donut (~0.2B), Pix2Struct-base (~0.3B), TrOCR (~0.3–0.6B) — strong but non-conversational, so they need task-specific harnessing rather than the shared VQA loop. **Out of <1B scope** (flagged for completeness): MonkeyOCR-pro-1.2B, Kosmos-2.5 (~1.3B), dots.ocr (~1.7B), Janus-Pro-1B (~1.5B, non-permissive license), Qwen2-VL-2B, Ovis2-2B, InternVL2.5/3.5-2B.

Architecture / pretraining / capability profiles for each model are in **Appendix A**.

## 2. Benchmark selection & design

Document understanding spans recognition (read the pixels), layout (where things are), and reasoning (combine fields, do arithmetic). No single dataset covers all three, so I use a **suite** plus a **custom probe**:

| Benchmark               | Source (VLMEvalKit / HF)                  | Primary metric    | What it probes                                                     | Why it's here                                |
|-------------------------|-------------------------------------------|-------------------|--------------------------------------------------------------------|----------------------------------------------|
| <b>DocVQA</b>           | lmms-lab/DocVQA (DocVQA, val)             | ANLS              | Printed text + layout + KIE on scanned business docs               | The canonical document-QA benchmark          |
| <b>InfoVQA</b>          | lmms-lab/DocVQA (InfographicVQA, val)     | ANLS              | Dense multi-element layout, text+chart fusion, numeric reasoning   | Hardest layout/reasoning; complements DocVQA |
| <b>ChartQA</b>          | lmms-lab/ChartQA (test; human/aug slices) | Relaxed acc       | Reading + arithmetic over plotted data                             | Structured-visual reasoning                  |
| <b>OCRBench</b>         | echo840/OCRBench                          | OCRBench (/ 1000) | Regular/irregular/handwritten/artistic text, KIE, handwritten math | Fine-grained recognition capability          |
| <b>Robustness probe</b> | derived from DocVQA (ours)                | ANLS retention    | Capture-quality + terminology stability                            | The "beyond accuracy" deployment axis        |

**Why validation splits.** DocVQA/InfoVQA **test** labels live behind an eval server; I use **val** so results are self-contained and reproducible. (Published model cards often quote test — a split mismatch I flag explicitly in the comparison table so numbers are not silently conflated.)

**The custom robustness probe (design decisions).** Real documents are phone photos, faxes, and re-compressed PDFs, and users phrase questions in domain jargon. Headline ANLS

on pristine scans hides this. The probe takes N DocVQA items and emits a **paired** set — a `clean` baseline plus controlled perturbations — so failures are attributable:

- **downscale** (low-DPI / small photo → small-text legibility)
- **jpeg** quality 18 (heavy re-compression artifacts)
- **blur** (out-of-focus capture)
- **rotate** 5° (skew / handheld tilt)
- **noise** (sensor / photocopier speckle)
- **term\_paraphrase** (image untouched; question rewritten with terse domain wording — "total"→"aggregate", "how much"→"what is the amount" — a deterministic, rule-based terminology stress test)

Robustness is reported as **retention** = **score(perturbed)** / **score(clean)** per family, which separates "accurate" from "accurate and stable".

### 3. Evaluation metrics — beyond accuracy

| Metric                      | Definition                                                             | Why it matters for documents                                                                                             |
|-----------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>ANLS</b>                 | 1 – normalised edit distance to best gold; 0 below 0.5                 | Official DocVQA/InfoVQA metric; tolerant of minor OCR/format noise (e.g. \$1,200 vs 1200 ) without rewarding near-misses |
| <b>Relaxed accuracy</b>     | Numeric within 5% rel. error, else exact match                         | Official ChartQA metric; the right notion of "correct" for read-off numbers                                              |
| <b>OCRBench score</b>       | Gold string $\subseteq$ prediction, summed /1000                       | Standard OCRBench scoring; isolates recognition from reasoning                                                           |
| <b>Calibration — ECE</b>    | $\Sigma$                                                               | $\text{acc}(\text{bin}) - \text{conf}(\text{bin})$                                                                       |
| <b>Robustness retention</b> | $\text{score}(\text{perturbed})/\text{score}(\text{clean})$ per family | Production inputs are degraded; retention predicts real-world accuracy better than clean ANLS                            |
| <b>Operational</b>          | answer-rate, load time, avg latency/sample                             | Edge viability: a model that's accurate but 10× slower may be unusable                                                   |

Confidence for ECE is the **mean token probability** of the generated answer, read from HF `generate(output_scores=True)` . Genuinely autoregressive backends (InternVL, SmoVLm, LLaVA-OV, PaddleOCR-VL) expose it; transcription-only models (GOT, Florence-2) do not, and are reported as `ECE = n/a` rather than with a fabricated number.

### 4. Experimental setup & reproducibility

- **Hardware.** Free **Google Colab / Kaggle T4 (16 GB)** — every model fits in fp16/bf16; no paid hardware required, as the task allows.
- **Decoding (held constant across all models for fairness).** Greedy ( `do_sample=False` , `num_beams=1` ), `max_new_tokens=256` , bf16. Greedy ⇒ deterministic ⇒ reproducible.
- **Preprocessing.** Each model uses **its own native, documented** image pipeline (InternVL dynamic tiling up to 12×448px; SmoVLm/LLaVA AnyRes; Florence/GOT task pipelines). Forcing a shared resize would unfairly cripple tiling-based models whose document edge comes from high-res tiling — so "consistency" here means consistent decoding + consistent scoring, not identical pixels. This choice is documented so it is auditable.

- **Determinism.** Fixed seeds; greedy decoding; pinned package versions ( `requirements.txt` ); predictions cached to `predictions.jsonl` so re-scoring never re-runs the model and runs are resumable.
- **Fairness controls.** Same prompt template per task; same sample subset ( `--limit` ) across models; identical metric code for all.

## 5. Software stack

| Tool                                | Role                                          | Why                                                                                                                                                                                                                                                                                                |
|-------------------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PyTorch + HF Transformers</b>    | Model loading/ inference                      | Every candidate ships HF weights; one API ( <code>AutoModel*</code> , <code>generate</code> ) covers them via thin adapters                                                                                                                                                                        |
| <b>HF Datasets</b>                  | Benchmark sourcing                            | Canonical hosting for DocVQA/InfoVQA/ChartQA/OCRBench                                                                                                                                                                                                                                              |
| <b>VLMEvalKit</b>                   | Cross-check / standard harness                | The task's reference harness; we mirror its dataset choices & metrics, and provide a thin wrapper. We add a self-contained pipeline on top because VLMEvalKit reports accuracy/ANLS but <b>not calibration or our robustness retention</b> — the metrics that make this analysis "beyond accuracy" |
| <b>PEFT (LoRA)</b>                  | Part-2 fine-tuning PoC                        | Parameter-efficient adaptation fits the sub-1B/edge story and a T4 budget                                                                                                                                                                                                                          |
| <b>Pillow / NumPy / torchvision</b> | Image I/O + perturbations + tiling transforms | Lightweight, ubiquitous                                                                                                                                                                                                                                                                            |

**Why a custom pipeline and VLMEvalKit.** VLMEvalKit is the right standard for headline accuracy and is referenced for cross-checking. But calibration (ECE) and the paired robustness probe are not in its metric set, and they are central to the document deployment story — hence the small, tested `docvlm_eval` package, whose single `evaluate.py` "loads any model, runs the benchmark, outputs per-model scores" exactly as the PoC requires.

## Part 2 — Knowledge gap & improvement strategy

### 1. Knowledge-gap analysis (evidence-based)

Using **published reference figures** for the sub-1B models (sources & caveats in the comparison table) and the structure of the benchmark suite, three gaps emerge. The improvement strategy targets the **best model, InternVL2.5-1B**, against its own weakest axis.

**Gap A — InfoVQA  $\ll$  DocVQA (the primary gap).** Across every model the InfoVQA score sits far below DocVQA (InternVL2.5-1B: **56.0 vs 84.8**; InternVL3-1B: 53.7 vs 81.9; LLaVA-OV-0.5B: 46.3 vs 73.7). InfoVQA is not harder to read — it is harder to reason over: dense infographics demand (i) parsing many spatially-scattered elements, (ii) fusing text with embedded charts/ icons, and (iii) multi-step numeric reasoning. The 0.5B language model is the bottleneck: recognition is solved, layout-grounded reasoning is not.

**Gap B — recognition  $\neq$  reasoning.** OCR specialists (GOT, Florence-2) are built for transcription and report **no** DocVQA/InfoVQA/OCRBench-VQA scores because they cannot answer questions; PaddleOCR-VL is evaluated only on parsing (OmniDocBench edit-distance),

not VQA. This confirms that pushing OCR quality alone will not close the document-QA gap — the deficit is in reasoning over recognised content, exactly where Gap A points.

**Gap C — reliability is unmeasured by leaderboards.** No public card reports **calibration** or **degraded-input robustness** for these models, yet both decide deployability. Small LMs are known to be over-confident; our pipeline measures ECE and per-perturbation retention to expose it. (These are produced by running the provided pipeline; they are the columns leaderboards omit.)

**Why InternVL2.5-1B is the target.** It is the strongest published small document model (DocVQA 84.8, OCRBench 785, ChartQA 75.9, InfoVQA 56.0). Its DocVQA/OCRBench/ChartQA are near-saturated for the size, so the **highest-leverage, lowest-risk** improvement is its worst axis — **InfoVQA / layout-grounded numeric reasoning** (Gap A) — while hardening reliability (Gap C).

## 2. Improvement strategy (concrete, justified)

A staged plan, each step tied to a gap and to literature, all runnable on a single T4 — and backed by the **LoRA fine-tuning scaffold already in this repo** (`src/ocr_ft`, `scripts/finetune_lora.py`).

**Step 1 — Targeted LoRA SFT on layout-reasoning data (attacks Gap A).** Parameter-efficient **LoRA** (Hu et al., 2021) / **QLoRA** (Dettmers et al., 2023) on the LM + projector, freezing InternViT. Train on a reasoning-heavy document mix: **InfographicVQA train**, **DocVQA train**, **ChartQA**, **Docmatix** (the doc instruction set behind SmolVLM), and **TabMWP/PlotQA-style** numeric items. Rationale: the recognition stack is already strong; LoRA cheaply re-weights the reasoning pathway toward multi-element/numeric questions without catastrophic forgetting, and fits a T4. Keep dynamic high-res tiling on — it is what lets a 0.5B LM see small infographic text.

**Step 2 — Reasoning distillation from a larger teacher (amplifies Gap A).** Generate **chain-of-thought rationales** for InfoVQA/ChartQA train questions with a larger teacher (InternVL-8B/26B or a frontier VLM) and fine-tune the 1B student on (image, question → rationale → answer). Sequence-level KD / rationale distillation transfers reasoning procedure, not just answers, and is well-suited to small students (Hinton et al., 2015; Hsieh et al., "Distilling step-by-step", 2023). This directly addresses the multi-step numeric reasoning that InfoVQA exposes.

**Step 3 — Robustness-aware augmentation (attacks Gap C / retention).** Augment SFT with the **same perturbations as the robustness probe** (downscale/jpeg/blur/ rotate/noise) and **terminology paraphrases**. Standard augmentation theory + the paired probe give a direct read on whether retention improves. Keep augmented and clean copies so clean accuracy is preserved.

**Step 4 — Calibration (attacks Gap C / ECE).** Post-hoc **temperature scaling** on a held-out doc set (Guo et al., 2017) — one parameter, no accuracy change, large ECE reduction — optionally with **confidence-aware decoding** so the deployed reader can abstain/route low-confidence fields. Re-measure ECE with the pipeline.

**Why this combination.** LoRA+QLoRA = T4-feasible and forgetting-safe; distillation injects the reasoning the small LM lacks; augmentation + temperature scaling convert clean-set wins into

deployable wins. The plan is surgical — it spends capacity on the one axis the evidence says is weak (InfoVQA/layout reasoning), not on already-saturated OCR.

### 3. Expected outcomes & measurement

| Axis                            | Baseline (published ref.) | Target after plan                | Measured by                |
|---------------------------------|---------------------------|----------------------------------|----------------------------|
| InfoVQA (val ANLS)              | ~56                       | <b>+6-10 ANLS</b>                | evaluate.py on InfoVQA     |
| ChartQA (relaxed)               | ~76                       | +2-4                             | evaluate.py on ChartQA     |
| DocVQA (val ANLS)               | ~85                       | <b>no regression</b> ( $\pm 1$ ) | evaluate.py on DocVQA      |
| Robustness worst-case retention | TBD (pipeline)            | <b>+0.1-0.2</b>                  | robustness probe retention |
| Calibration (ECE)               | TBD (pipeline)            | <b>halved</b>                    | ECE in summary.json        |

Expected magnitudes are deliberately modest and InfoVQA-focused because that is where the headroom is; the success criterion is **InfoVQA up materially with DocVQA/OCR Bench held**, plus measurable reliability gains. Everything is verified by re-running the same pipeline, so before/after is apples-to-apples.

---

## Appendix A — Model profiles

- **InternVL2.5-1B / InternVL3-1B** (OpenGVLab/InternVL2\_5-1B, OpenGVLab/InternVL3-1B, MIT). InternViT-300M vision encoder + Qwen2.5-0.5B LM via an MLP pixel-shuffle projector, with **dynamic high-resolution tiling** (up to 12× 448px crops + thumbnail). Trained on large multimodal corpora with heavy OCR/document/chart data; v3 adds a native multimodal pretraining recipe. Strong OCR/doc/chart for the size.
- **SmolVLM-256M / 500M** (HuggingFaceTB/SmolVLM-\*-Instruct, Apache-2.0). SigLIP vision encoder + SmolLM2 LM in the Lefids3 framework, with aggressive pixel-shuffle token compression for on-device use; training mix includes **Docmatix** (document instructions).
- **LLaVA-OneVision-0.5B** (llava-hf/llava-onevision-qwen2-0.5b-ov-hf, Apache-2.0). SigLIP-SO400M + Qwen2-0.5B with AnyRes tiling; strong general single-image/multi-image/video VLM, not document-specialised.
- **GOT-OCR2.0** (stepfun-ai/GOT-OCR-2.0-hf, Apache-2.0). ~80M ViT encoder + Qwen-0.5B decoder trained end-to-end for **transcription** (plain/formatted/markdown). No free-form question interface → OCR-specialist baseline.
- **Florence-2-large** (microsoft/Florence-2-large, MIT). DaViT encoder + BART-style enc-dec trained on FLD-5B with a **task-token** interface (<OCR>, <OD>, ...). No <DocVQA> token → driven with <OCR>; specialist, not conversational.
- **PaddleOCR-VL-0.9B** (PaddlePaddle/PaddleOCR-VL). NaViT-style dynamic-resolution encoder
- ERNIE-4.5-0.3B LM, purpose-built for full-page **document parsing** (text/table/formula/chart, reading order) across 100+ languages; evaluated by its authors on parsing (OmniDocBench edit-distance), not VQA.



## Appendix B — Selected references

---

- Mathew et al. DocVQA: A Dataset for VQA on Document Images. WACV 2021.
- Mathew et al. InfographicVQA. WACV 2022.
- Masry et al. ChartQA. ACL Findings 2022.
- Liu et al. OCRBench: On the Hidden Mystery of OCR in Large Multimodal Models. 2023.
- Chen et al. InternVL / InternVL 2.5 (arXiv:2412.05271) / InternVL3 (arXiv:2504.10479).
- Marafioti et al. SmolVLM. arXiv:2504.05299.
- Li et al. LLaVA-OneVision. arXiv:2408.03326.
- Wei et al. General OCR Theory (GOT-OCR2.0). arXiv:2409.01704.
- Xiao et al. Florence-2. arXiv:2311.06242 (CVPR 2024).
- PaddleOCR-VL Technical Report. arXiv:2510.14528.
- Hu et al. LoRA. arXiv:2106.09685. Dettmers et al. QLoRA. arXiv:2305.14314.
- Hinton et al. Distilling the Knowledge in a Neural Network. 2015. Hsieh et al. Distilling Step-by-Step. ACL 2023.
- Guo et al. On Calibration of Modern Neural Networks. ICML 2017.

Published scores quoted in this report are self-reported by model authors (papers/cards), evaluated via VLMEvalKit/OpenCompass; split conventions (val vs test) and scale conventions (OCRBench /1000 vs /100) differ between sources and are flagged in the comparison table.